

In [7]: `import numpy as np`

```
def f(x):
    return 2 * x**3 + 4 * x**2 - 6 * x + 1

a, b = 1, 5
h = 0.5
n = int((b - a) / h)
x = np.linspace(a, b, n + 1)
y = f(x)

integral_approx = (h / 2) * (f(a)+f(b) + 2 * np.sum(y[1:-1]))
print(f"Approximate integral (Trapezoidal Rule): {integral_approx:.4f}")

def F(x):
    return (x**4 / 2) + (4 * x**3 / 3) - 3 * x**2 + x

exact = F(b) - F(a)
print(f"Exact integral: {exact:.4f}")

error = abs(integral_approx - exact) / exact * 100
print(f"Percentage error: {error:.4f}%")
```

Approximate integral (Trapezoidal Rule): 413.0000
Exact integral: 409.3333
Percentage error: 0.8958%

In []: